Third Section. The roots of algebraic equations.

§ 30. Concept of roots. Multiple roots.

After, in the first two sections, the algebraic magnitudes had been considered more from the formal side, where the matter was identical transformations of letter-expressions in which the letters throughout could be conceived as symbols for variable magnitudes, the numerical magnitudes now come more into the foreground.

Here, in accordance with what was fixed in the Introduction, by numbers we understand real or imaginary magnitudes of the form $a + bi$, and for illustration we represent the real magnitudes by points of a straight line and the imaginary magnitudes by points of a plane. By the absolute value of an imaginary magnitude $a + bi$ we understand $\sqrt{a^2 + b^2}$ and denote it, following Weierstrass, by

$$|a + bi|.$$

Let now

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

be an integral function of x, in which the coefficients are arbitrary real or imaginary numbers, and the first, a_0, is assumed different from zero. If α is a number which, when put for x, makes the function $f(x)$ equal to zero, that is, which satisfies the condition $f(\alpha) = 0$, then α is called a root of the equation

$$f(x) = 0.$$

We also say briefly that α is a root of $f(x)$. By § 4, $f(x)$ may then be divided by $x - \alpha$ without remainder, so that one may put

$$f(x) = (x - \alpha)f_1(x), \tag{2}$$

where $f_1(x)$ is only of the $(n - 1)$st degree and has the same first coefficient a_0 as $f(x)$, thus

$$f_1(x) = a_0x^{n-1} + \cdots .$$

The coefficients of $f_1(x)$ are given in § 4, (6).

Every root of $f_1(x)$ is therefore at the same time a root of $f(x)$; and conversely every root of $f(x)$ is either $= \alpha$ or is a root of $f_1(x)$. If one root α of $f(x)$ is known, then the task of finding the remaining ones is reduced to the solution of an equation of the $(n - 1)$st degree.

The aim of the considerations of this section consists in proving that every function $f(x)$ of the nth degree has at least one root. This theorem is called the fundamental theorem of algebra. First we draw from (2) the important consequence:

I. An equation of the nth degree cannot have more than n roots.

For if $f(x)$ had more than n roots, then $f_1(x)$, which is only of the $(n - 1)$st degree, would have more than $n - 1$ roots. But an equation of the first degree has not more than one root, whence the correctness of our theorem follows by complete induction.

Sometimes one also gives it the expression:

II. If a function $f(x)$ of the nth degree has more than n roots, then all its coefficients must be zero.

If β is a root of $f_1(x)$, then one may likewise put

$$f_1(x) = (x - \beta)f_2(x),$$

where $f_2(x) = a_0x^{n-2} + \cdots$ is only of the $(n - 2)$nd degree. Thus, if one assumes that each of the functions obtained by this division, $f_1(x), f_2(x), \ldots$, has at least one root, one finally obtains

$$f(x) = a_0(x - \alpha)(x - \beta) \cdots (x - \nu), \tag{3}$$

and we can therefore also express the theorem which we earlier described as the aim of our considerations as follows. It is to be proved:

A function of the nth degree can be decomposed into n linear factors.

The magnitudes $\alpha, \beta, \ldots, \nu$ are then all roots of $f(x)$, and their number is therefore n. It is, however, not excluded that the same number occurs more than once among $\alpha, \beta, \ldots, \nu$. Then $f(x)$ would have fewer than n roots, while the theorem would still remain that $f(x)$ is decomposable into n linear factors. To establish agreement, it has been agreed that, if $x - \alpha$ goes into $f(x)$ several times, α is to be counted several times among the roots, and hence that one speaks of simple, double, triple, etc. roots.

By the concept of the derived functions [§ 13, (2)], if α is an arbitrary magnitude,

$$f(x) = f(\alpha) + (x-\alpha)f'(\alpha) + \frac{(x-\alpha)^2}{1\cdot 2}f''(\alpha) + \frac{(x-\alpha)^3}{1\cdot 2\cdot 3}f'''(\alpha) + \cdots . \tag{4}$$

If it is now again assumed that $f(\alpha)$ vanishes, then there results

$$f_1(x) = f'(\alpha) + \frac{x-\alpha}{1\cdot 2}f''(\alpha) + \frac{(x-\alpha)^2}{1\cdot 2\cdot 3}f'''(\alpha) + \cdots ,$$

and hence

$$f_1(\alpha) = f'(\alpha).$$

Thus α is a double root of $f(x)$ when $f'(\alpha)$ vanishes simultaneously with $f(\alpha)$. Formula (4) also shows that only under this assumption is $f(x)$ divisible by $(x-\alpha)^2$. This conclusion may be carried further and leads to the theorem:

III. The necessary and sufficient condition that α be an m-fold root of $f(x)$ is that $f(\alpha), f'(\alpha), f''(\alpha), \ldots, f^{(m-1)}(\alpha)$ vanish, while $f^{(m)}(\alpha)$ does not vanish.

§ 31. Continuity of integral functions.

If, as before,

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_n$$

is an integral function of x, we first prove the following theorem:

IV. $f(x)$ becomes infinitely large at the same time as x.

This means that, if C is an arbitrarily given positive number, the positive number R can be so chosen that

$$|f(x)| > C$$

as soon as the absolute value of x becomes greater than R; or, expressed geometrically: one can describe in the x-plane a circle around the origin such that outside this circle the absolute value of $f(x)$ no longer falls below C, however large C is assumed. The theorem is evident for the case of a simple power x^n; for if r is the absolute value of x, then r^n is the absolute value of x^n, and r^n, as soon as r has become greater than 1, is greater than r and hence grows with r into the infinite.

To prove the theorem in general, let us denote the absolute values of $x, a_0, a_1, a_2, \ldots, a_n$ by $r, c_0, c_1, c_2, \ldots, c_n$. Then

$$|f(x)| = r^n \left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| , \tag{2}$$

and, since by the theorems proved in the Introduction the absolute value of a sum cannot be smaller than the difference and cannot be greater than the sum of the absolute values of the summands,

$$\left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| \geq c_0 - \left| \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| \geq c_0 - \frac{c_1}{r} - \frac{c_2}{r^2} - \cdots - \frac{c_n}{r^n}.$$

Since $c_0, c_1, \ldots, c_n$ are fixed constants, one can choose R so large that, as soon as $r > R$,

$$\frac{c_1}{r} + \frac{c_2}{r^2} + \cdots + \frac{c_n}{r^n}$$

becomes arbitrarily small, for example smaller than $\frac{1}{2}c_0$. Then

$$\left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| > \tfrac{1}{2}c_0,$$

and hence by (2)

$$|f(x)| > \tfrac{1}{2}c_0 R^n,$$

so that, if one assumes

$$R^n > \frac{2C}{c_0}, \tag{3}$$

one has

$$|f(x)| > C.$$

To this is attached the theorem:

V. $f(x)$ is a continuous function of x.

By this the following is meant. If the absolute value of x is smaller than an arbitrary finite magnitude R, then, however small the positive magnitude ω is assumed, one can find a positive magnitude ε such that

$$|f(x+h) - f(x)| < \omega \tag{4}$$

as long as the absolute value ρ of h remains smaller than ε, and indeed in such a way that ε depends only on the choice of ω, not on x.

By § 13, (2),

$$f(x+h) - f(x) = hf'(x) + \frac{h^2}{1 \cdot 2}f''(x) + \cdots + h^n a_0. \tag{5}$$

If now we assume the absolute value of h smaller than 1, then the absolute value of the magnitude in parentheses

$$f'(x) + \frac{h}{1 \cdot 2}f''(x) + \cdots + h^{n-1}a_0 \tag{6}$$

is smaller than a certain non-zero number K which one obtains if in (6) one replaces h by 1, the coefficients $a_0, a_1, \ldots, a_{n-1}$ by their absolute values and x by R, because thereby every individual term of the sum (6) is replaced by its absolute value or by a larger number. If one chooses ε so small that

$$\varepsilon K < \omega,$$

then by (5) the requirement (4) is fulfilled as long as $\rho < \varepsilon$.

We can give the theorem of the continuity of the function $f(x)$ a somewhat different expression, which is important for what follows. By the theorem that the absolute value of a sum of two terms lies, in magnitude, between the sum and the difference of the absolute values of the terms (cf. Introduction, p. 19), we obtain from (5), if we put

$$\varphi(h) = hf'(x) + \frac{h^2}{1 \cdot 2}f''(x) + \cdots + h^n a_0,$$

$$|f(x)| - |\varphi(h)| < |f(x+h)| < |f(x)| + |\varphi(h)|,$$

or

$$-|\varphi(h)| < |f(x+h)| - |f(x)| < |\varphi(h)|.$$

Now, if $\rho < \varepsilon$, then $|\varphi(h)| < \omega$, and we can therefore also say:

VI. The absolute value of the difference

$$|f(x+h)| - |f(x)|$$

is smaller than an arbitrarily given magnitude ω as soon as the absolute value of h is smaller than a sufficiently small ε.

From this we can still conclude, if we put $x = y + iz$ and assume h either real or purely imaginary, that $|f(y+zi)|$ is, for fixed z, a continuous function of y, and for fixed y, a continuous function of z.

If the coefficients $a_0, a_1, \ldots, a_n$ and the variable x are real, then x may be chosen, in absolute value, so large that the sign of $f(x)$ agrees with the sign of the first term $a_0 x^n$; thus, for positive a_0 and even n, it becomes positive, while for odd n it becomes negative for negative x and positive for positive x.

For in

$$f(x) = x^n \left(a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right)$$

one can choose x so large in absolute value that the absolute value of

$$\frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

falls below that of a_0, and therefore the expression in parentheses

$$a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

has the sign of a_0, which it then retains for every absolutely larger x.

§ 32. Change of sign of $f(x)$. Roots of equations of odd degree and of pure equations.

On the basis of the theorems of the preceding paragraph we can prove the following important theorem.

VIII. If the coefficients of $f(x)$ are real and there exist two real values a, b of x such that $f(a)$ and $f(b)$ have opposite signs, then between a and b there is at least one root ξ of the equation $f(x) = 0$.

We shall assume that $a < b$ and that $f(a)$ is negative, $f(b)$ positive. We divide the numbers between a and b into two parts $\mathfrak{A}$ and $\mathfrak{B}$ as follows: a number α belongs to $\mathfrak{A}$ if, between α and a, including the endpoints, there is no x for which $f(x)$ becomes positive; and a number β belongs to $\mathfrak{B}$ if between β and a there is at least one value of x for which $f(x)$ becomes positive. A third possibility is plainly not possible, and every number between a and b is therefore placed in $\mathfrak{A}$ or in $\mathfrak{B}$. We also count the numbers smaller than a as belonging to $\mathfrak{A}$, and those greater than b as belonging to $\mathfrak{B}$.

If α belongs to $\mathfrak{A}$, then every x smaller than α also belongs to $\mathfrak{A}$; and if β belongs to $\mathfrak{B}$, every larger x belongs to $\mathfrak{B}$. Thus every β is greater than every α, and the two numerical domains $\mathfrak{A}, \mathfrak{B}$ form a cut (Introduction, p. 5), generated by a number ξ, such that every number α smaller than ξ belongs to $\mathfrak{A}$, and every number β greater than ξ belongs to $\mathfrak{B}$.

If now, for some number x_0, the value of $f(x_0)$ is negative or positive, then by V of the preceding paragraph a positive magnitude ε can be so determined that for every x between $x_0 - \varepsilon$ and $x_0 + \varepsilon$ the values of $f(x)$ are always negative or always positive. It follows from this that $f(\xi)$ can be neither negative nor positive and must therefore be zero. For if $f(\xi)$ were negative, then $f(x)$ would also be negative between ξ and $\xi + \varepsilon$; these values would therefore belong to $\mathfrak{A}$, although they are greater than ξ. If $f(\xi)$ were positive, then $f(x)$ would be positive as long as x lay between $\xi - \varepsilon$

and ξ; these values, although smaller than ξ, would therefore belong to $\mathfrak{B}$. Both alternatives are impossible by the definition of ξ. The same conclusion may be drawn in quite the same way when $f(a)$ is positive and $f(b)$ negative, and so our theorem is proved.

From this we draw two important consequences. If $f(\xi)$ vanishes, then an interval

$$\xi - \varepsilon < x < \xi + \varepsilon$$

can be so determined that $f(x)$ does not vanish in this interval except for $x = \xi$. Then four cases are possible with respect to the signs of $f(x)$ in the two intervals $\xi - \varepsilon < x < \xi$ and $\xi < x < \xi + \varepsilon$, which are put together in the following scheme:

	$\xi - \varepsilon < x < \xi$	$\xi < x < \xi + \varepsilon$
1.	$+$	$-$
2.	$-$	$+$
3.	$+$	$+$
4.	$-$	$-$

If $f^{(\nu)}(x)$ is the first among the derivatives of $f(x)$ which is different from zero for $x = \xi$, then the formula [§ 13, (2)]

$$f(\xi + h) = \frac{h^\nu}{\Pi(\nu)} f^{(\nu)}(\xi) + \cdots$$

shows that these four cases are distinguished by the following signs:

$$\begin{array}{lll} 1. & \nu \text{ odd}, & f^{(\nu)}(\xi) < 0, \\ 2. & \nu \text{ odd}, & f^{(\nu)}(\xi) > 0, \\ 3. & \nu \text{ even}, & f^{(\nu)}(\xi) > 0, \\ 4. & \nu \text{ even}, & f^{(\nu)}(\xi) < 0. \end{array}$$

We say that in the first case $f(x)$ passes decreasing through zero, and in the second increasing through zero. In case 3, $f(\xi) = 0$ is a minimum, in case 4 a maximum.

In particular, if $f'(\xi)$ is different from zero, so that ξ is a simple root, then the positive or negative sign of $f'(\xi)$ is the characteristic mark for the increase or decrease of the function $f(x)$ as x passes through ξ.

With these aids we still prove two further important theorems.

IX. The equation

$$x^n - a = 0$$

has, when a is a positive number, one and only one positive root.

For if we put

$$f(x) = x^n - a, \tag{1}$$

then by § 31, VII, $f(x)$ is positive for sufficiently large positive x, and negative for $x = 0$; hence there is a positive value of x, denoted by $\sqrt[n]{a}$, for which $f(x)$ vanishes. That there can be only one such value follows from the fact that x^n, as long as x remains positive, increases continually with x.

If n is odd, then also for negative a there exists one and only one negative root of (1), which is proved in the same way.

X. An equation of odd degree with real coefficients always has at least one real root.

For by theorem VII, $f(x)$, if the degree is odd, can become both positive and negative; hence, by VIII, $f(x) = 0$ has a root.

Finally we prove:

XI. A pure equation always has a root.

By a pure equation we understand an equation of the form

$$x^n - a = 0, \tag{2}$$

where a is an arbitrary complex magnitude. That this equation always has a root for real positive a has already been proved above; and for $a = 0$ it is plainly satisfied by $x = 0$.

The general proof can be divided into two parts; it is sufficient to assume first $n = 2$, and then n odd. For if the magnitude $\sqrt{a}$ and $\sqrt[\nu]{a}$, or $\sqrt[\nu]{\alpha}$ for every α, exists, then also the existence of $\sqrt[n]{a}$ for every arbitrary n is secured. It would even suffice to prove the existence of $\sqrt[n]{a}$ for the case in which n is a prime number. But from this no particular advantage is at first to be drawn.

We denote the complex magnitude a by $b + ci$ and put, since x will in general also be complex,

$$x = y + iz.$$

Then we first have to show that

$$(y + iz)^2 = b + ic, \tag{3}$$

whatever real values b, c may have, can be satisfied by real y, z. Equation (3), however, is fulfilled if

$$y^2 - z^2 = b, \qquad 2yz = c, \tag{4}$$

and hence

$$(y^2 + z^2)^2 = b^2 + c^2,$$

therefore

$$y^2 + z^2 = \sqrt{b^2 + c^2}. \tag{5}$$

By IX there always exists a positive square root $\sqrt{b^2 + c^2}$. From (4) and (5) it now follows that

$$y^2 = \frac{b + \sqrt{b^2 + c^2}}{2}, \qquad z^2 = \frac{-b + \sqrt{b^2 + c^2}}{2},$$

and the two magnitudes $\pm b + \sqrt{b^2 + c^2}$ are plainly positive, since $b^2 + c^2 > b^2$, hence also $\sqrt{b^2 + c^2} > \sqrt{b^2}$. Thus we obtain

$$y = \pm\sqrt{\frac{b + \sqrt{b^2 + c^2}}{2}}, \qquad z = \pm\sqrt{\frac{-b + \sqrt{b^2 + c^2}}{2}},$$

but one sign is determined by the other through the condition

$$2yz = c;$$

one of the two signs remains arbitrary, so that we obtain not merely one, but two opposite solutions of (3).

Further let n be odd; we take the equation to be solved in the form

$$(y + iz)^n = b + ci. \tag{6}$$

If $c = 0$, so that a is real, then we can put $z = 0$ and by IX obtain a real value of x. But if we assume c different from zero, then from (6) (Introduction, p. 19) there follows

$$(y - iz)^n = b - ci. \tag{7}$$

Multiplying the two equations (6), (7) by one another gives

$$(y^2 + z^2)^n = b^2 + c^2,$$

and hence, by IX,

$$y^2 + z^2 = \sqrt[n]{b^2 + c^2}, \tag{8}$$

which determines the absolute value of x. Furthermore, from (6) and (7) one obtains

$$\varphi(y,z) = (y-iz)^n(b+ic) - (y+iz)^n(b-ci) = 0. \tag{9}$$

Now $\varphi(y,z)$ is an integral homogeneous function of the nth degree in the two variables y,z, and indeed with real coefficients, since replacing i by $-i$ changes nothing in $\varphi(y,z)$. If it is ordered by descending powers of y, the first term is $2ciy^n$, hence, by assumption, different from zero. Putting

$$y = \lambda z,$$

we obtain from (9)

$$\varphi(\lambda, 1) = 0, \tag{10}$$

therefore an equation of the nth degree for λ, which by X certainly has a root. Once λ is determined, it follows from (8) that

$$y = \lambda\sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}, \qquad z = \sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}. \tag{11}$$

Through this the equations (8), (9) are in fact satisfied; from (9), however, follows

$$\frac{(y+iz)^n}{b+ic} = \frac{(y-iz)^n}{b-ci},$$

and from (8) that the common value of these two fractions is $+1$. Therefore we have

$$(y+iz)^n = \pm(b+ci), \qquad (y-iz)^n = \pm(b-ci),$$

and thus the sign of the square root in (11) can be so determined that equations (6) and (7) are fulfilled.

§ 33. Solution of pure equations by trigonometric functions.

The solvability of a pure equation can be shown far more completely and simply if one presupposes the trigonometric functions and their simplest properties as known. These functions are, it is true, foreign to algebra proper, and for that reason it is more satisfactory to answer the questions of principle without their aid, as has been done in the preceding and will also be done later. For application and for a more convenient visualization, however, we shall not dispense with this aid.

If, when b,c are real numbers, we put

$$b = r\cos\varphi, \qquad c = r\sin\varphi, \tag{1}$$

then, if r is assumed positive, the angle φ is thereby determined up to a multiple of 2π. It will be completely determined if we fix an interval of length 2π in which it is to lie, for example

$$-\pi < \varphi \leq \pi.$$

In geometrical interpretation r,φ are the polar coordinates of the point whose rectangular coordinates are b,c. The absolute value of the complex magnitude $a = b+ci$ is r, and we shall call φ the phase of this complex magnitude.

The product of two complex magnitudes

$$a = r(\cos\varphi + i\sin\varphi), \qquad a' = r'(\cos\varphi' + i\sin\varphi')$$

is obtained in the form

$$aa' = rr'[\cos(\varphi+\varphi') + i\sin(\varphi+\varphi')], \tag{3}$$

and from this, if n is an arbitrary positive whole number, or even a negative one, one obtains the theorem named after Moivre,

$$(\cos\varphi + i\sin\varphi)^n = \cos n\varphi + i\sin n\varphi, \tag{4}$$

which has led to regarding the magnitude $\cos\varphi + i\sin\varphi$ as an exponential function with imaginary exponent, and hence to setting

$$\cos\varphi + i\sin\varphi = e^{i\varphi}.$$

Thereby formulas (3) and (4) take the form

$$e^{i\varphi}e^{i\varphi'} = e^{i(\varphi+\varphi')}, \qquad (e^{i\varphi})^n = e^{in\varphi}. \tag{5}$$

If accordingly we put

$$a = r(\cos\varphi + i\sin\varphi) \tag{6}$$

and by $\sqrt[n]{r}$ understand the positive value existing and completely determined by § 32, IX, whose nth power is r, then

$$x = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right) \tag{7}$$

is a magnitude whose nth power is a, hence a root of the equation

$$x^n = a, \tag{8}$$

where n is understood to be an arbitrary positive whole number. But every one of the values

$$x_k = \sqrt[n]{r}\left(\cos\frac{\varphi+2k\pi}{n} + i\sin\frac{\varphi+2k\pi}{n}\right) \tag{9}$$

satisfies the same requirement, when k is an arbitrary integer. In (9), however, not more than n different values are contained, namely those obtained by putting

$$k = 0, 1, 2, \ldots, n-1; \tag{10}$$

for if k is increased by a multiple of n, the value of (9) is not changed, while the n values (10) give only different values of x, because they have different phases.

Equation (8) consequently has not merely one, but n different roots. The geometrical images of the values x_k all lie on a circle with radius $\sqrt[n]{r}$, and are separated from one another by the angle $2\pi/n$. The first of these points is obtained by dividing the given angle φ into n equal parts. The radius $\sqrt[n]{r}$ is smaller or larger than r according as r is smaller or larger than 1. The adjacent figure shows these relations for $n = 3$ under the assumption $r > 1$.

One can obtain the different values x_k by multiplying the first of them,

$$x_0 = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right),$$

by the different values of

$$\xi_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}. \tag{11}$$

The magnitudes ξ_k,

$$\xi_k^n = 1, \tag{12}$$

are roots of the equation $x^n = 1$ and are called the nth roots of unity. There are n, and only n, different values of them; for odd n only one is real, namely for $k = 0$, and for even n two are real, namely for $k = 0$ and $k = n/2$. The geometrical images of the nth roots of unity are the vertices of a regular n-gon inscribed in the circle of radius 1. Their algebraic determination is the subject of the theory of cyclotomy. If we denote by ε the value of ξ_k for $k = 1$, thus put

$$\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n},$$

then, by Moivre's theorem,

$$\xi_k = \varepsilon^k,$$

so that all nth roots of unity are represented as powers of one among them.